INDUSTRY EXPERT VALIDATION

Findings Summary and Sign-Off Document

Predictive Anemia Screening and Healthcare Resource Planning Using Full Blood Count Data in Sri Lankan Healthcare Institutions

Purpose of This Document

This document has been prepared by the student consultancy team for IT3081 (Statistical Modelling) to summarize the key findings of our analysis and to formally record the professional feedback of an independent industry expert, as required by Task 11 of the group assignment (Industry Expert Validation). Please review the summary below, discuss it with the consultancy team if useful, and complete the accompanying Feedback Form. Your signature at the end of this document confirms that the consultation took place and that the summarized findings were presented to you accurately.

Consultancy Engagement Details

Project title	Predictive Anemia Screening and Healthcare Resource Planning Using FBC Data in Sri Lankan Healthcare Institutions
Module	IT3081 — Statistical Modelling, Group Assignment (20%)
Consultancy theme	Public Health Indicator Analysis
Dataset	222,042-record Full Blood Count dataset, four anonymized Sri Lankan healthcare institutions (2014–2023), sourced from Zenodo and restructured by the consultancy team
Consultation date	_____
Consultation format	☐ In person ☐ Video call ☐ Phone call ☐ Written correspondence (email)

Summary of Key Findings Presented to the Expert

1. Descriptive and inferential findings

- Overall anemia prevalence across all cases was 43.6%, with substantial variation by subgroup.
- A significant age × sex interaction was found (two-way ANOVA, $p < 0.001$): women show much higher anemia prevalence in reproductive years (51.6% at ages 18–39 vs. 9.5% in men), but this gap narrows and reverses in old age (80+: 73.7% women vs. 80.5% men).
- Anemia prevalence differed significantly by institution (chi-square test, $p < 0.001$), though the effect size was small (Cramer's $V = 0.093$) and concentrated mainly in one institution.

2. Predictive modelling findings

- A logistic regression model (age, sex, and non-hemoglobin red-cell indices, excluding institution) achieved 94.4% accuracy and an AUC of approximately 0.99 in predicting anemia status on held-out test data.
- Regularized alternatives (Ridge, LASSO, Elastic Net) were compared to address measured multicollinearity among red-cell indices; Elastic Net is recommended for operational deployment.

3. Bayesian decision-making findings

- Using an expected-loss framework with an illustrative cost ratio (missing a true case costs 5× a false alarm), shifting the classification threshold from the conventional 0.5 to approximately 0.15 reduced total expected error cost by 44.5% on test data.

4. Proposed innovation

- An Anemia Early Warning and Resource Allocation Framework (AEWRAF) combining a risk-scoring engine, a calibrated decision layer, clinical triage alerts, and an institutional resource-planning dashboard - built entirely from data already collected in routine care, requiring no new equipment or tests.

Areas on Which Expert Feedback Is Requested

The consultancy team requests the expert's professional judgement on the following four areas, corresponding to the accompanying Feedback Form:

- Relevance of the problem - is anemia screening/resource planning a genuine priority for Sri Lankan healthcare institutions?
- Practicality of recommendations - are the specific recommendations (screening model, decision threshold, dashboard) realistic given real clinical and operational constraints?
- Implementation feasibility - what practical barriers would this framework face if deployment were attempted?
- Organizational impact - what impact, positive or negative, would this framework likely have on the organization if implemented?

Acknowledgement and Sign-Off

By signing below, the expert confirms that the consultancy team presented the above findings, that the expert had the opportunity to ask questions and provide professional feedback, and that the accompanying Feedback Form reflects their genuine professional opinion.

Expert details

Full name: _____

Professional title / role: _____

Organization: _____

Years of relevant experience: _____

Contact email / phone: _____

Sign-off

Signature: _____

Date: _____

Consultancy team member(s) present: _____